**Sequence 15**

# CONSOLIDATION AND TRANSITION TO YEAR 12

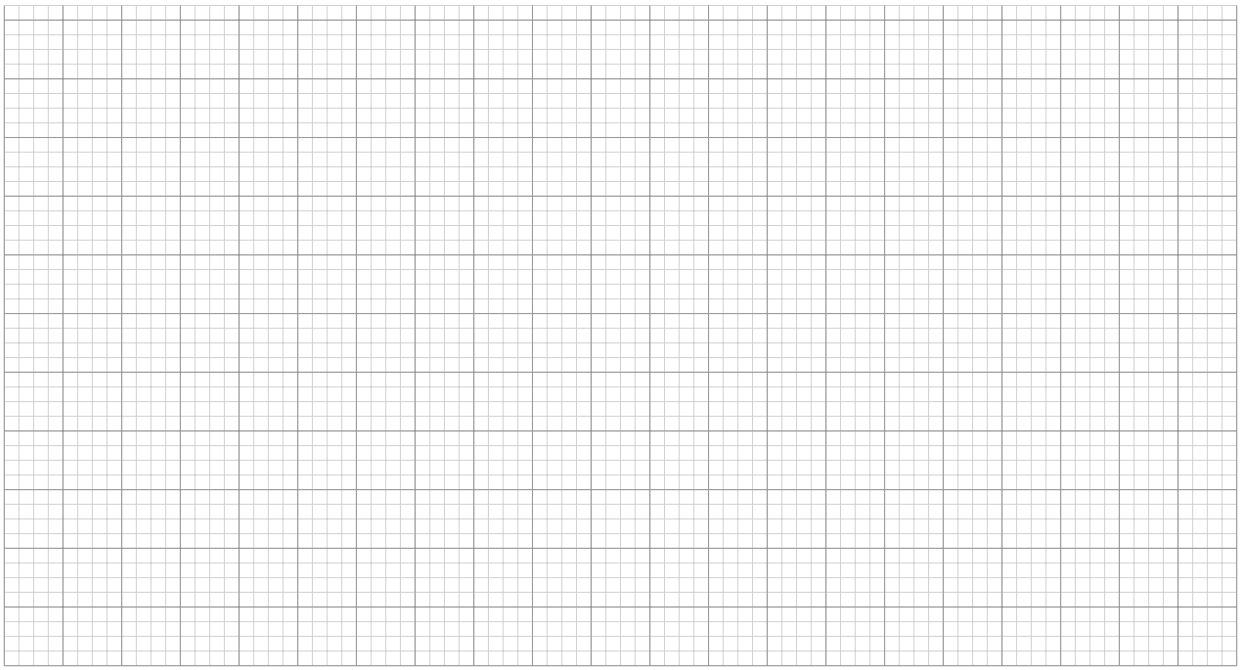
25 May–11 June 2027

## I Review of calculation, functions and geometry

### Example 15.1: Annual review – part A

1. Reduce $168/252$ to lowest terms; calculate $2^{-3} + 1/4$.
2. Factorise $x^2 - 6x + 9$, then solve $(x-3)^2 = 4$.
3. Solve $\dfrac{x-2}{x+1} \geq 0$.
4. For $f(x) = 4 - (x-1)^2$ on $[-2;3]$, give the maximum and the abscissae where $f(x) = 0$.
5. With $A(0;1)$, $B(4;3)$, $C(6;4)$, prove collinearity and give an equation of the line.

**Solution:**

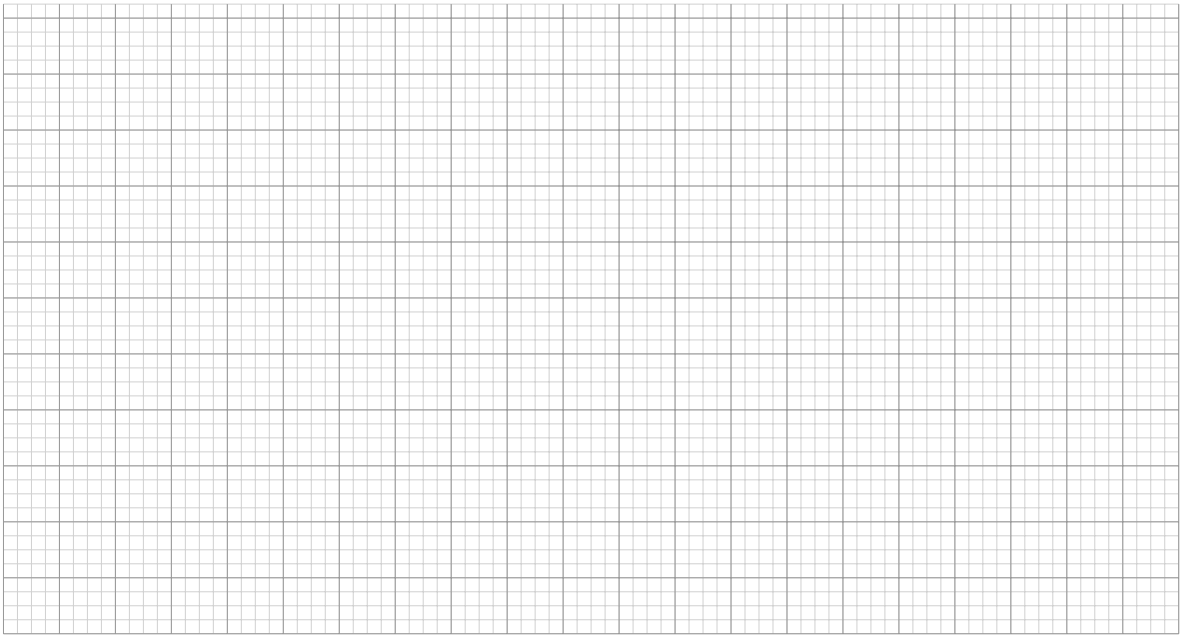


## II Review of data, probability and modelling

**Example 15.2: Annual review – part B**

1. Calculate the mean and median of $2; 4; 4; 6; 9$. Determine $Q_1, Q_3$.
2. A price drops by $15\,\%$, then rises by $20\,\%$. Give the overall rate and the rate that then restores the initial price.
3. In a fictional population of 100 people, 40 play sport ($S$), 30 come by bike ($V$), 18 do both. Construct the two-way table and calculate $P_S(V)$, $P_V(S)$ for an equally likely draw.
4. For a rectangle of perimeter 24 m, express the area as a function of one dimension $x$ and prove its maximum.

**Solution:**



## III Choosing a strategy and preparing what comes next

**Example 15.3: A program to explain**

For a given function $f$, explain the following program. What does it return with $f(x) = x^2$?

```
def tableau(f):
    valeurs = []
    for x in range(-3, 4):
        valeurs.append((x, f(x)))
    return valeurs
```

**Solution:**

**Six skills to mobilise**

**Search**: try things out, organise leads. **Model**: choose variables, domain and assumptions. **Represent**: move from text to table, figure or graph. **Reason**: justify and distinguish conjecture from proof. **Calculate**: choose a form and check the operations. **Communicate**: explain clearly with suitable notation and units.

**Example 15.4: End-of-year portfolio**

Choose a successful piece of work and a corrected one. For each, identify the skills used, explain one method and state a personal goal for the following year. Then redo an exercise matching this goal without looking at the correction.

**Solution:**